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Shift-share analysis: further examination of models for the description of economic change

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Abstract

Shift-share is a widely-used technique for the analysis of regional economies. As a methodology, shift-share is comprised of traditional accounting-based models, Analysis of Variance models, and information-theoretic models. The purpose of this paper is to present and demonstrate the usefulness of two probabilistic forms of shift-share models. These highly flexible variance partitioning methods are but one example of the broader class of models used in the analysis of aggregate, tabular data within planning, geography and regional science. Further, probabilistic shift-share provides a major advance over traditional accounting-based methods because it allows the researcher to quantitatively test hypotheses about changes in employment or value added by region or sector. Also, the casting of shift-share analysis in this light offers proof of the adequacy of these models. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Why investigate shift-share again?

Shift-share is a widely-used technique for the analysis of regional economies. The shift-share problem involves partitioning, say, employment change, into that due to national trends, industrial sector trends and local conditions. A survey of the literature indicates that shift-share analysis continues to be popular among planners, geographers and regional scientists. It has been utilized in structuralist political economy [1–3], retail analysis [4], migration analysis [5,6],

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and neoclassical analyses of regional growth [7,8]. Additionally, policy-makers who often have need of quick, inexpensive analysis tools that are neither mathematically complex nor data intensive also utilize shift-share extensively. Despite its widespread use, criticisms of the technique abound. Reservations center on such issues as temporal, spatial, and industrial aggregation [9,10,11], theoretical content [8,11–14], and predictive capabilities [11,14–17].

Traditionally, shift-share analysis has utilized accounting identities, but the shift-share problem also can be thought of as partitioning of variation in a three dimensional contingency table having the dimensions industrial sector, spatial unit, and year. As such, shift-share analysis is a special case of a very general set of descriptive statistical models of aggregate tabular data that play a central role in the analysis of geographic and regional issues.

The purpose of this paper is to present and demonstrate the usefulness of probabilistic forms of shift-share analysis. The paper also includes an empirical comparison of traditional accounting-based and probabilistic models. Results indicate that probabilistic shift-share provides a major advance over traditional accounting-based methods because it allows the researcher to quantitatively test hypotheses about changes in employment or value added by region or sector. Also, the casting of shift-share analysis in this light offers proof of the adequacy of these models. Choice among probabilistic shift-share models appears to be somewhat problematic, with both Analysis of Variance-based (ANOVA) models and information-theoretic models having strengths and weaknesses. While information-theoretic models require minimal prior data manipulation and allow customization of the model to account for available information, they are difficult to interpret and may not provide superior goodness-of-fit when compared with ANOVA-based models.

In the following section, traditional accounting-based shift-share is summarized. This is followed by a review of ANOVA-based shift-share in the third section of the paper. These reviews include minor extensions to the basic shift-share framework, for example the Arcelus extension and dynamic shift-share, but not the Rigby–Anderson [18] or Haynes–Dinc [19] extensions. Next, traditional accounting-based and ANOVA-based shift-share are compared. This empirical comparison is followed by the introduction of information-theoretic shift-share models, including the Arcelus extension, dynamic shift-share, and an additional mixed-scale form to illustrate the flexibility of the information-theoretic approach. An empirical comparison of the ANOVA and information-theoretic models follows. The final portion of the paper summarizes the study findings and provides conclusions.

2. Accounting-based shift-share

In order to make sense of what is to follow, it is first necessary to briefly review traditional accounting-based shift-share. The National Growth Rate version of traditional, accounting-based shift-share [20–22] partitions change in a regional economic indicator, for example, employment, into components representing national share, n_i^r, proportional shift, s_i^r, (an industry mix effect), and differential shift, d_i^r (a local competitive effect):

$$c_i^r = n_i^r + s_i^r + d_i^r \tag{1}$$

where c_i^r is change in sector i in region r. In considering employment, each component is

defined as follows:

$$n_i^r = \acute{E}_i^r g^n \tag{1a}$$

$$s_i^r = \acute{E}_i^r (g_i^n - g^n) \tag{1b}$$

$$d_i^r = \acute{E}_i^r (g_i^r - g_i^n) \tag{1c}$$

where $\acute{E}_i^r$ is employment in sector i in region r in the end year, g^n is the growth rate for total employment, g_i^n is the employment growth rate in sector i, and g_i^r is the regional growth rate for sector i. Growth rates are determined as follows:

$$g^n = (\Sigma_{ri}\acute{E}_i^r - \Sigma_{ri}E_i^r)/\Sigma_{ri}E_i^r \tag{1d}$$

$$g_i^n = (\Sigma_r\acute{E}_i^r - \Sigma_r E_i^r)/\Sigma_r E_i^r \tag{1e}$$

$$g_i^r = (\acute{E}_i^r - E_i^r)/E_i^r \tag{1f}$$

where E_i^r is employment in sector i in region r in the base year.

Use of shift-share in the form presented in Eq. (1) is today rather rare, thanks to two more recent improvements in the basic form. The first of these, the Arcelus extension [23,24], includes terms for national share, n_i^r, proportional shift, s_i^r, regional shift, r_i^r, and differential shift, d_i^r:

$$c_i^r = n_i^r + s_i^r + r_i^r + d_i^r \tag{2}$$

where c_i^r, n_i^r and s_i^r are defined as in Eq. (1) and:

$$r_i^r = \acute{E}_i^r (g^r - g^n) \tag{2a}$$

$$d_i^r = \acute{E}_i^r\left[(g_i^r - g^r) - (g_i^n - g^n)\right] \tag{2b}$$

where:

$$g^r = (\Sigma_i\acute{E}_i^r - \Sigma_i E_i^r/\Sigma_i E_i^r) \tag{2c}$$

Dynamic shift-share [9] is directed at a resolution of problems arising because of changing industrial mix and regional employment growth. The method is an extension of Thirwall's [25] suggestion that the study period be sub-divided into two or more subperiods to reduce the severity of changes in industrial mix on the results of the analysis (for examples, see [26–30]). The dynamic approach has the advantage of adjusting annually for change in industrial mix, continuously updating regional employment, and using annual growth rates. In so doing, the dynamic approach provides a more accurate allocation of regional employment change among effects and allows for the identification of unusual years and years of economic transition.

Dynamic shift-share provides a solution to the problem of changing industrial mix as well as a correct estimate of national versus regional growth. To produce a dynamic model, Eqs. (1) or (2) can be computed for all adjacent years. The researcher may then present average annual change or sum annual change to obtain total national share, sectoral share, and regional share across the study period:

$$\Sigma_k c^r_{ik} = \Sigma_k n^r_{ik} + \Sigma_k s^r_{ik} + \Sigma_k d^r_{ik} \tag{3}$$

where $k = 1,\ldots,T$ is a sequence of adjacent years. The summation can be for a period of any length.

3. ANOVA-based shift-share models

Berzeg [30] provides a statistical basis for shift-share in terms of Analysis of Variance (ANOVA). In particular, he shows that the shift-share identity can be formalized as the linear model:

$$g^r_i = a + B_i + e^r_i \tag{4}$$

where the national growth rate, g^n, is estimated as the model's intercept, a, B_i is an estimate of $(g^n_i - g^n)$, and error term e^r_i provides an estimate of $(g^r_i - g^n_i)$. Assuming a normal distribution for the e^r_i, the ratio of the parameters to their standard errors will be distributed as Student's t and traditional measures of goodness-of-fit are appropriate.

ANOVA-based shift-share is easily adapted to either the Arcelus extension or to the dynamic context. A probabilistic model of the Arcelus extension has the form (compare [31]):

$$g^r_i = a + B_i + G_r + e^r_i \tag{5}$$

where terms are defined as in Eq. (4), but where $G_r = (g^r - g^n)$ and

$$e^r_i = (g^r - g^n) - (g^n_i - g^n).$$

As before, assuming a normal distribution for the e^r_i, the ratio of the parameters to their standard errors will be distributed as Student's t and traditional measures of goodness-of-fit are appropriate.

Adaptation of the model to the dynamic context might include the simple application of models Eqs. (4) or (5) to each of a succession of years. Alternatively, a dynamic model might involve the addition of another dimension to the analysis, i.e.,

$$g^r_i = a + B_i + G_r + V_k + e^r_i. \tag{6}$$

In the static case, $a = g^n$, $B_i = g^n_i - g^n$ and $e^r_i = g^r_i - g^n_i$. In Eq. (6), $a = \Sigma_k g^n_k/k$, $B_i = \Sigma_k (g^n_{ik} - g^n_k)/k$ and $e^r_{ik} = \Sigma_k (g^r_{ik} - g^n_i)/k$ where k denotes the number of pairs of adjacent years.

4. The relationship between traditional and ANOVA shift-share

The analytical relationship between the various forms of the shift-share model will be discussed using models (1) and (4), while a more complex comparison will be undertaken empirically. Models (1) and (4) represent the same formulation, the simplest of all shift-share models. Yet, while models (1) and (4) are mathematically equivalent, model (4) when calibrated will not produce parameters that are identical to those provided by (1) for two reasons. First, Eq. (4) is valued in terms of growth rates while Eq. (1) is valued in terms of employment. Thus, terms in Eqs. (1) and (4) will differ by E_i^r. Second, in Eq. (4), terms e_i^r are heteroscedastic; hence, a is not precisely g^n and B_i is not precisely $g^n - g_i^n$. Berzeg [31,32] shows that proper choice of weights and use of weighted least squares will yield identical estimates in Eqs. (1) and (4). In particular, identical estimates are obtained if it is assumed that $E[e_i^{r2}] = (w_i^r)^{-1}\sigma^2$ instead of the usual assumption that $E[e_i^{r2}] = \sigma^2$ where $w_i^r = E_i^r / \Sigma_{ri} E_i^r$. Berzeg's method and an alternative calibration procedure devised by Patterson [33] are compared in the Appendix to this paper. Berzeg's method is shown to be more accurate and less computationally expensive than that proposed by Patterson.

The empirical comparison of traditional and ANOVA-based shift-share utilizes data compiled by the Bureau of Labor Statistics (BLS) of the US Department of Labor for 1939–90 as part of their *Establishment Survey of Employment, Hours, and Earnings*. The data include entries by BLS industry division and five digit Standard Industrial Code (SIC) for all US states and possessions. Data on annual employment are comprised of averages of monthly data surveyed on the twelfth day of each month [34]. Here, these data are aggregated sectorally into the 9 BLS divisions and spatially into the nine Census divisions.

Results of the empirical comparison of models (1) and (4) appear in Table 1. Years 1970 and 1980 are used in this simple comparison. Values in column 1 of the table are generated by the traditional shift-share model (1). Column 2 of the table presents results for model (4). A

Table 1
Comparison of results from simple models

Parameter	Model (1)	Model (4)
Intercept	0.4721	0.2117
Mining	0.0393	0.2997
Construction	0.3118	0.5722
Durable manufacturing	0.4905	0.7509
Nondurable manufacturing	0.2896	0.5500
Transportation and public utilities	−0.0241	0.2363
Wholesale and retail trade	−0.0283	0.2321
F.I.R.E.	0.0735	0.3339
Services	−0.0646	0.1958
Government	−0.2604	Aliased[a]
R^2	N.A.	0.1836

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.

scatterplot (Fig. 1) of model residuals indicates conformity to the assumption of a normal distribution.

Model (4) is a statistical model, not an accounting model; hence, degrees of freedom must be considered. Shift-share, from a statistical perspective, is tabular having m rows (sectors) but only $m-1$ degrees of freedom. As a result, the parameter associated with one sector cannot be estimated while the remaining parameters are determined only up to an additive constant. The values for proportional shift in Eq. (1) and the parameters obtained for Eq. (4) are thus related in the following way. The intercept of Eq. (4) equals the national share component of Eq. (1) plus the proportional shift component associated with government employment, the variable in Eq. (4) which cannot be estimated. From Table 1, $0.2117-(-0.2604)=0.2117+0.2604=0.4721$. In the same way, the remaining parameters in Eq. (4) equal the corresponding proportional shift components in Eq. (1) minus the proportional shift component for government employment. It is also evident from the table that the model provides a poor explanation of the changes in US employment patterns over the study period. Despite this, some information can be gleaned from model (4). The proportional shift parameters associated with construction, durable manufacturing, and nondurable manufacturing are significantly different from the proportional shift parameter associated with government employment (whose parameter is zero by definition).

Results of the empirical comparison of the Arcelus extended models (2) and (5) appear in Table 2. A scatterplot of model (5) residuals (Fig. 2) again demonstrated the reasonableness of

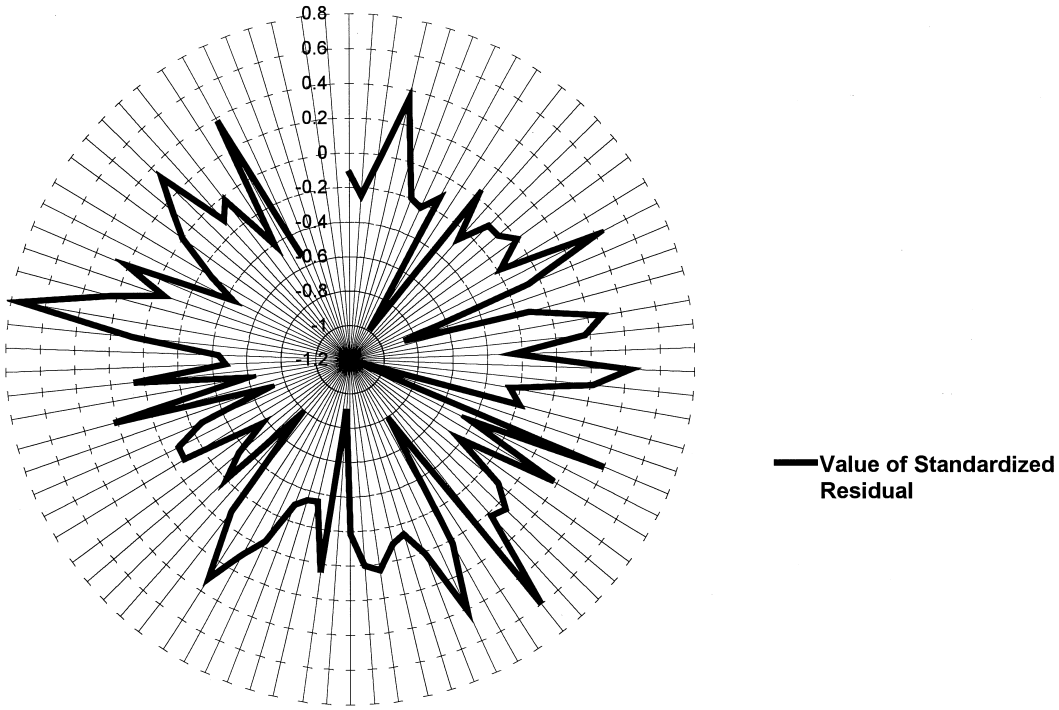


Fig. 1. Standardized residual plot for model (4).

the assumption of a normal distribution. Values in column 1 of the table are generated by the traditional accounting-based model (2). Column 2 of the table presents results for model (5).

In this instance, the parameter associated with one sector and one region cannot be estimated and the remaining parameters are determined only up to an additive constant. The intercept of Eq. (5) should therefore equal the national share component of Eq. (2) plus the proportional shift component associated with government employment and the region shift parameter associated with the Pacific region — the two parameters in Eq. (5) that cannot be estimated. The remaining proportional parameters in Eq. (5) should equal the corresponding proportional shift component in Eq. (2) minus the proportional shift component for government employment, while the remaining regional shift parameters in Eq. (5) should equal the corresponding regional shift parameters in Eq. (2) minus the regional shift parameter associated with the Pacific region.

However, this is not the case and some deviation exists between models (2) and (5). In model (5), for example, goodness-of-fit is improved over that of model (4). Only the proportional shift parameter for durable manufacturing is significantly different from the proportional shift parameter of government employment (whose parameter is zero by definition).

Results of the empirical comparison of dynamic models (3) and (6) appear in Table 3. Values in column 1 of the table are generated by the conventional shift-share model (3). Column 2 of the table presents results for model (6). Again, the parameter associated with one sector and one region cannot be estimated and the remaining parameters are determined only

Table 2
Comparison of results from Arcelus extended models

Parameter	Model (2)	Model (5)
Intercept	0.4721	0.4184
Mining	0.0393	0.2306
Construction	0.3118	0.5308
Durable manufacturing	0.4905	0.6572
Nondurable manufacturing	0.2896	0.4703
Transportation and public utilities	−0.0241	0.1132
Wholesale and retail trade	−0.0283	0.0952
F.I.R.E.	0.0735	0.3372
Services	−0.0646	0.1324
Government	−0.2604	Aliased[a]
New England	−0.0851	−0.2175
Middle Atlantic	0.0149	−0.1459
South Atlantic	−0.2325	−0.3101
East South Central	0.1387	−0.0237
East North Central	0.2206	−0.0577
West North Central	0.0579	−0.2512
West South Central	−0.1780	−0.2842
Mountain	0.2069	0.0837
Pacific	0.1051	Aliased[a]
R^2	N.A.	0.2284

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.

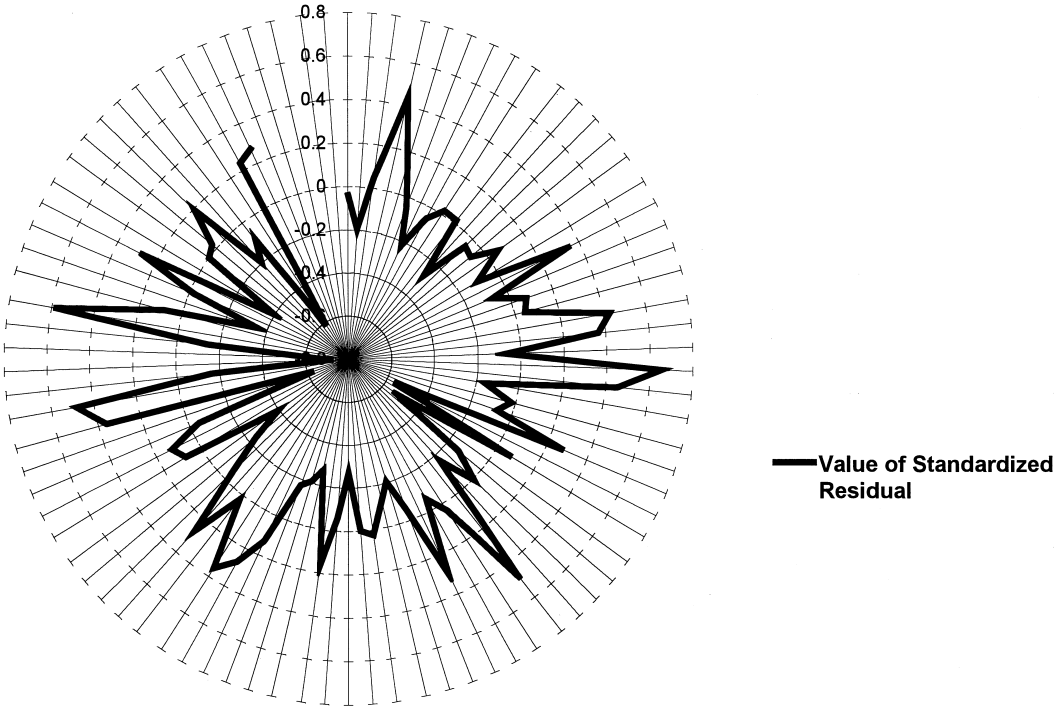


Fig. 2. Standardized residual plot for model (5).

up to an additive constant. Although poor model fit makes interpretation extremely problematic, there again appears to be considerable deviation in the results generated by models (3) and (6). However, if the parameter values of government employment and the Pacific region are subtracted from the intercept of Eq. (6), the resulting value is a close approximation of the national share effect in Eq. (3). Experimentation shows that the bulk of variation in these models is vested in the temporal parameter (V_k in Eq. (6)) and that the dynamic shift-share strategy proposed by Barff and Knight [9] is the best way of capturing this.

5. Information-theoretic shift-share models

Previous use and methodological innovation in the realm of shift-share has taken place within the traditional accounting and ANOVA-based techniques. A principal difference between the bulk of previous research on shift-share and the research reported here is a reliance on information-theoretic methods. In order to illustrate the efficacy of these methods, it is necessary to illustrate two things. First, that information-theoretic methods lead to models that are at least as diverse as the those obtained under the traditional accounting-based and ANOVA-based formats. Second, it must be shown that they are superior performers based on standard goodness-of-fit measures.

Shift-share analyses based on information theory rely on the information gain measure of

Kullback and Leibler [35]:

$$I(P:Q) = \Sigma_{ri} p_i^r \ln(p_i^r/q_i^r) \tag{7}$$

where, q_i^r represents an element of a prior probability distribution and p_i^r represents an element of a posterior probability distribution. The measure I(P:Q) thus measures the information gained from observing distribution P given distribution Q. Within the context of shift-share analysis, $p_i^r = \acute{E}_i^r/\Sigma_{ri}\acute{E}_i^r$ and $q_i^r = E_i^r/\Sigma_{ri}E_i^r$. Construction of a variety of shift-share models is possible by considering the general problem:

$$\min I(P:Q) = \Sigma_{ri} p_i^r \ln(p_i^r/q_i^r)\ p_i^r \tag{8}$$

subject to:

1. $\Sigma_{ri} p_i^r f^t(x_i^{rt}) = x^t$; for all t

2. $\Sigma\Sigma_{ri} p_i^r = 1$

3. $p_i^r \geqslant 0$; for all i and j

where $f^t(x_i^{rt})$ represents some function in t of observed values x_i^r and x^t represents some

Table 3
Comparison of results from dynamic models

Parameter	Model (3)	Model (6)
Intercept	1.2446	0.6367
Mining	−0.3853	0.1334
Construction	−0.1685	0.0751
Durable manufacturing	−0.0205	0.3176
Nondurable manufacturing	−0.3403	0.0863
Transportation and public utilities	−0.5246	−0.0763
Wholesale and retail trade	−0.3083	0.2189
F.I.R.E.	0.1980	0.0946
Services	0.6107	0.3520
Government	0.0998	Aliased[a]
New England	0.5689	0.6244
Middle Atlantic	1.0037	0.8224
South Atlantic	1.4318	0.8535
East South Central	0.3811	0.6742
East North Central	−0.2255	0.4096
West North Central	−0.5601	0.0654
West South Central	−0.8353	−0.1981
Mountain	−0.7044	−0.0322
Pacific	−0.6945	Aliased[a]
R^2	N.A.	0.0111

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.

exogenously specified limiting value. Direct solution of Eq. (8) is possible, but it is more convenient to solve the dual of Eq. (8) which is an unconstrained geometric programming problem [36,37]:

$$\max \mathrm{F}(\lambda^t) = \Sigma_{ri} q_i^r \exp\left[\Sigma_t \lambda^t f^t(x_i^{rt})\right] + \Sigma_t \lambda^t x^t + \lambda^0 \tag{9}$$

where λ^t and λ^0 are unconstrained dual variables [38]. This approach to the solution of Eq. (9) is not generally employed in planning, regional science and geography. Rather, planners, regional scientists and geographers tend to exploit two other properties of Eqs. (8) and (9).

First, when the constraints in Eq. (8) are linear equalities, programs (8) and (9) lead directly to a model of quantities p_i^r from the application of Lagrangian methods:

$$\ln p_i^r = \ln q_i^r - \lambda^0 - \Sigma_t \lambda^t f^t(x_i^{rt}). \tag{10}$$

Second, quantities p_i^r, λ^t, and λ^0 that result from the solution of Eqs. (8), (9) or (10) are asymptotic maximum likelihood estimators [39]. As a result, planners, regional scientists and geographers tend to estimate model (10) using maximum likelihood methods [40,41]. Use of the maximum likelihood approach ensures asymptotic normality of the model parameters [42]. This, in turn, simplifies hypothesis testing and assessment of goodness-of-fit. The ratio of parameters to their standard errors is asymptotically normal and goodness-of-fit is measured as deviance which is asymptotic chi-square distributed.

Information gain has been used to derive both shift-share measures of spatial concentration [43,44] and shift-share models based on minimum discrimination information (MDI) [45–49]. To arrive at the information-based model proposed by Theil and Gosh [45], first form the MDI primal:

$$\min \mathrm{I(P{:}Q)} = \Sigma_{ri} p_i^r \ln(p_i^r / q_i^r) \tag{11}$$

subject to:

1. $\Sigma_r p_i^r = \mathrm{o}_i$; for all i

2. $p_i^r \geqslant 0$; for all r and i

where o_i is the frequency of employment in sector i and other terms are as previously defined. In this case, application of Lagrangian methods results in the model:

$$p_i^r = q_i^r \exp(\alpha + \mathrm{o}_i) + f^r_i. \tag{12}$$

This model can in turn be rewritten as the dynamic model [50]:

$$\acute{E}_i^r = E_i^r \exp(A + \mathrm{O}_i) + F_i^r \tag{13}$$

where O_i is total employment in industry i, A is the intercept and F_i^r is the error term. Suitable rearrangement yields:

$$\ln(\acute{E}_i^r/E_i^r) = A + \mathrm{O}_i + \ln(F_i^r), \tag{14}$$

which is easily calibrated using Generalized Linear Modeling methods [51–59].

First, consider the Arcelus extension. This model can be formulated by addition of the constraint

$$\Sigma_i p_i^r = d^r; \text{ for all } r \tag{15}$$

to Eq. (11) above, resulting in the model:

$$p_i^r = q_i^r \exp(\alpha + \mathrm{o}_i + d^r) + f_i^r \tag{16}$$

or,

$$\acute{E}_i^r = E_i^r \exp(A + \mathrm{O}_i + D^r) + F_i^r \tag{17}$$

which may be calibrated as:

$$\ln(\acute{E}_i^r/E_i^r) = A + \mathrm{O}_i + D^r + \ln(F_i^r) \tag{18}$$

where D^r is total regional employment respective of industry.

Models (14) and (18) can be extended to be dynamic in either of two ways. First, they may be sequentially applied and the results summed or averaged (see [9]). Second, just as the Arcelus extension includes a dimension representing region, a time dimension (denoted $k = 1,\ldots,T$) can be added to the analysis:

$$\ln(\acute{E}_i^r/E_i^r) = A + \mathrm{O}_i + D^r + T_k + \ln(F_i^r) \tag{19}$$

where T_k is total employment at time k respective of industry or region.

The consideration of shift-share within an information-theoretic context also clarifies further extensions of the model. The probabilistic shift-share models derived to this point have consisted of categorical variables arranged in tabular form. This view of shift-share creates an easily demonstrable practical problem. Consider an Arcelus-extended shift-share analysis. If, for example, concern lies with the performance of a county within Maryland relative to the Maryland economy, then all counties of Maryland must be included in the data (see [31]). By the same token, however, if concern lies with the performance of a county within Maryland relative to the US economy, then all counties of the US must be included in the analysis! This occurs because, in the probabilistic models, tests of significance of parameters rely on the concepts of variation about a mean, a concept quite irrelevant to the traditional accounting-based models. This does, however, potentially limit the practical use of the probabilistic methods to either small regions or to high levels of spatial aggregation. A possible solution is to use a model composed of a mixture of continuous and categorical variables. One may think of regional-level employment change data, say, those relating to the state of Maryland, as being a subset of the larger, national employment change data. Information on national share and national proportional shift, if they are to be included in an analysis of employment change in Maryland, must be drawn from the margins of the larger national table, not from the margins of the smaller Maryland table. In this sense, national share and national proportional

shift data are "outside of", or exogenous to, the data on employment change in Maryland. Within the probabilistic shift-share format, continuous variables are thus introduced to account for these exogenous data (see [60]).

Given these considerations, define the frequency of national employment in industry i as n_i. Using information theory, the MDI primal is:

$$\min \mathrm{I(P:Q)} = \Sigma_{ri} p_i^r \ln\left(p_i^r / q_i^r\right) \tag{20}$$

subject to:

1. $\Sigma_r p_i^r \ln(n_i) = n_i$; for all i

2. $\Sigma_r p_i^r = \mathrm{o}_i$; for all i

3. $p_i^r \geqslant 0$; for all r and i.

Application of Lagrangian methods results in the model:

$$p_i^r = q_i^r n_i \exp(\alpha + \mathrm{o}_i) + f_i^r. \tag{21}$$

This model can, in turn, be rewritten as:

$$\acute{E}_i^r = E_i^r . N_i . \exp(A + \mathrm{O}_i) + F_i^r, \tag{22}$$

which is calibrated as:

$$\ln(\acute{E}_i^r / E_i^r) = \ln(N_i) + A + \mathrm{O}_i + \ln(F_i^r). \tag{23}$$

A similar solution for ANOVA-based shift-share would be:

$$g_i^r = a + n_i + B_i + e_i^r, \tag{24}$$

where n_i is defined similar to B_i, but at the national scale, and all terms are as previously defined.

6. Empirical comparison of the information-theoretic approach with ANOVA-based methods

The previous section was designed to demonstrate the flexibility that the information-theoretic approach to shift-share affords. This section of the paper is devoted to comparing results of similarly formulated ANOVA and information theoretic models. Models (4) and (14) should yield identical results if Eq. (14) is calibrated using weights $(w_i^r)^{-1}$ and it is assumed that errors $\ln(F_i^r)$ are normally distributed [32,61] since $g_i^r = \acute{E}_i^r / E_i^r - 1$ and $-1 \leq g_i^r \leq \infty$ as $-\infty \leq \ln(\acute{E}_i^r / E_i^r) \leq \infty$.

Results of the empirical comparison of models (4) and (14) appear in Table 4. The years 1970 and 1980 again are used. Values in column 1 of the table are generated by the ANOVA model (4), while column 2 of the table presents results for model (14).

In both models, parameters are determined only up to an additive constant. There is no precise relationship between the parameters of models (14) and (4), although clearly a precise relationship should exist. In particular, if we denote model parameters of Eq. (4) as B_i and model parameters of Eq. (14) as O_i, then $B_i = \exp(O_i - 1)$. The lack of a precise relationship between the three models appears to stem from the necessity to generalize the $\mathbf{X}^T\mathbf{X}$ matrix to force nonsingularity during model calibration. Clearly, however, model (14) provides superior goodness-of-fit and greater discrimination. In the case of model (14), the intercept is significantly different from zero and the parameters associated with employment in all sectors are significantly different from that associated with employment in government. This finding corresponds to the noticeable slowing of employment growth in the governmental sector after the 1960s.

Results of the empirical comparison of Arcelus-extended models (5) and (18) appear in Table 5. Values in column 1 of the table are generated by the ANOVA model (4). Column 2 of the table presents results for information-theoretic model (18). In this instance, a parameter associated with one industry and one region cannot be estimated.

Again, no precise relationship exists between the parameters of model (18) and those of model (5). Model (18) appears to produce superior results. This is evidenced by the model's superior performance both in terms of goodness-of-fit and discrimination with respect to the parameters. In model (18), the intercept is significantly different from zero and the proportional shift parameters associated with employment in all sectors except wholesale and retail trade are significantly greater than that for government employment. All parameters except those associated with employment change in the Mountain and East South Central regions are significantly less than those associated with the Pacific region. These results from the shift-share analysis again correspond to what is known about the 1970–80 period. In particular, this period was one of relatively slow growth in government employment. Geographically, growth was centered in the southern and western regions of the United States.

Results of the empirical comparison of dynamic models (6) and (19) appear in Table 6.

Table 4
Comparison of results from simple models

Parameter	Model (4)	Model (13)
Intercept	0.2117	0.4038
Mining	0.2997	0.2247
Construction	0.5722	0.3625
Durable manufacturing	0.7509	0.4909
Nondurable manufacturing	0.5500	0.3748
Transportation and public utilities	0.2363	0.1786
Wholesale and retail trade	0.2321	0.1837
F.I.R.E.	0.3339	0.2136
Services	0.1958	0.1522
Government	Aliased[a]	Aliased[a]
R^2	0.1836	0.5163[b]

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.
[b] Denotes an R^2-type measure referred to elsewhere [60] as Percentage Null Model Deviance (PNMD).

Values in column 1 of the table are generated by the ANOVA model (6). Column 2 of the table presents results for model (19). Again, the parameter associated with one sector and one region cannot be estimated and the remaining parameters are determined only up to an additive constant. As before, no precise relationship exists between the parameters of models (6) and (19). Also, in this instance, model fit is equally dismal for the information-theoretic model (19) as for the ANOVA-based model (6).

Results of the mixed categorical and continuous variable model analysis appear in Table 7. These models are calibrated with less information than the previous models. Here, information on all sectors in the West North Central, Mountain and Pacific regions have been deleted from the data set. This is necessary so that the continuous variables do not retain redundant information. This also better simulates conditions as described earlier.

Values in column 1 of the table are generated by ANOVA-based model (24). In this instance, the parameters associated with the last two sectors cannot be estimated and the remaining sectoral parameters are determined only up to an additive constant. If the continuous, exogenous variable were functioning perfectly, the intercept of (24) would equal the national share component of (1) plus the proportional shift components associated with service and government employment. Ideally, the remaining proportional parameters in (24) should equal the corresponding proportional shift component in Eq. (1) minus the proportional shift components for service and government employment. Not surprisingly, the

Table 5
Comparison of results from Arcelus extended models

Parameter	Model (5)	Model (18)
Intercept	0.4184	0.3513
Mining	0.2306	0.3291
Construction	0.5308	0.4320
Durable manufacturing	0.6572	0.3164
Nondurable manufacturing	0.4703	0.0947
Transportation and public utilities	0.1132	0.0895
Wholesale and retail trade	0.0952	0.0895
F.I.R.E.	0.3372	0.2140
Services	0.1324	0.1027
Government	Aliased[a]	Aliased[a]
New England	−0.2175	−0.1323
Middle Atlantic	−0.1459	−0.0727
South Atlantic	−0.3101	−0.2123
East South Central	−0.0237	0.0014
East North Central	−0.0577	−0.0450
West North Central	−0.2512	−0.1332
West South Central	−0.2842	−0.1862
Mountain	0.0837	0.0704
Pacific	Aliased[a]	Aliased[a]
R^2	0.2284	0.6601[b]

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.
[b] Denotes an R^2-type measure referred to elsewhere [60] as Percentage Null Model Deviance (PNMD).

Table 6
Comparison of results from dynamic models

Parameter	Model (6)	Model (19)
Intercept	0.6367	0.6075
Mining	0.1334	0.0411
Construction	0.0751	0.0452
Durable manufacturing	0.3176	0.0877
Nondurable manufacturing	0.0863	0.1004
Transportation and public utilities	−0.0763	−0.0150
Wholesale and retail trade	0.2189	0.1089
F.I.R.E.	0.0946	−0.0703
Services	0.3520	0.0940
Government	Aliased[a]	Aliased[a]
New England	0.6244	0.1101
Middle Atlantic	0.8224	0.0827
South Atlantic	0.8535	0.0710
East South Central	0.6742	0.0069
East North Central	0.4096	0.0764
West North Central	0.0654	−0.1107
West South Central	−0.1981	−0.1922
Mountain	−0.0322	−0.0167
Pacific	Aliased[a]	Aliased[a]
R^2	0.0111	0.0258[b]

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.
[b] Denotes an R^2-type measure referred to elsewhere [60] as Percentage Null Model Deviance (PNMD).

Table 7
Results for the mixed models

Parameter	Model (24)	Model (23)
Intercept	0.5197	0.4937
National	−1.5047	−9.301e-07
Mining	0.1365	−0.0209
Construction	0.2815	0.1101
Durable manufacturing	0.6503	0.3441
Nondurable manufacturing	0.2642	0.1289
Transportation and public utilities	−0.0418	−0.1226
Wholesale and retail trade	0.2508	0.1782
F.I.R.E.	0.1400	−0.0326
Services	Aliased[a]	Aliased[a]
Government	Aliased[a]	Aliased[a]
R^2	0.1987	0.5162[b]

[a] Aliased parameters cannot be determined due to insufficient degrees of freedom.
[b] Denotes an R^2-type measure referred to elsewhere [60] as Percentage Null Model Deviance (PNMD).

exogenous variable is not functioning perfectly; hence, this is not the case. Rather, some deviation exists between models (1) and (24). However, model goodness-of-fit of Eq. (24) exceeds that of Eq. (1). This is unexpected since, theoretically, Eq. (24) is calibrated using less information than is the case for Eq. (1). This may be an artifact of the poor fit of model (1). Only the parameter associated with durable manufacturing significantly exceeds those associated with service and government employment (whose parameters are zero by definition).

Column 2 of the table presents results for model (23). If the continuous, exogenous variable were functioning perfectly, models (23) and (13) would produce identical results. As was the case for the ANOVA-based model, the exogenous variable is not functioning perfectly. Further, model goodness-of-fit of Eq. (13) exceeds that of Eq. (23). Since Eq. (23) is calibrated using less information, this is entirely expected. The parameters associated with durable manufacturing and wholesale and retail trade significantly exceed those associated with service and government employment (again, whose parameters are zero by definition).

7. Conclusions and implications of the research

Shift-share is a relatively old ex-post analysis technique that measures the ends of the process of change rather than the variables that are the agents of change. The technique additionally has been criticized because of its assumptions concerning the linearity of regional economic dynamics and its lack of ability to handle regional variation.

These objections, while containing some truth, ignore three realities. First, recent improvements such as the Arcelus extension and dynamic shift-share have lessened the technique's reliance on the assumed long-term linearity of regional dynamics. In so doing, they have created models much more able to measure regional variation. Indeed, the thrust of this paper has been to illustrate that probabilistic forms of shift-share models (and, particularly, information theory-based models) are members of a highly flexible class of variance-partitioning methods, akin to the family of spatial interaction models delineated by Wilson [41]. This implies that shift-share is but one example of the broader class of models used in the analysis of aggregate tabular data within planning, geography and regional science. More generally, probabilistic shift-share provides a major advance over traditional accounting-based methods because it allows the researcher to quantitatively test hypotheses about changes in employment or value added by region or sector [61].

Second, the construction of causal models of regional economic development presupposes that, as researchers, we have a clearly defined theory of regional economic development. This would appear not to be the case. The last decade has been a period of extreme tumult among both regional economic development theoreticians and regional economies. While variance-partitioning models, such as shift-share, do not explain economic phenomenon, they do allow us to allocate variation among competing alternatives (national sectoral shift versus regional shift, independent of sector), and thus bring us closer to explanation.

Also, probabilistic shift-share has a number of implications for the validation of theories in geography and regional science that are best demonstrated by example. Consider the ongoing debate on the relative importance of the global economy versus the local economy in determining employment patterns (classics include [62,63]). Shift-share models contain two

components, national share and proportional shift, that might be termed "global", and another term, differential shift, that might be termed "local" in that it approximates, somewhat crudely, regional or local uniqueness. Suppose that models such as Eqs. (1), (4), and (14) are employed in the analysis of employment trends and it is found that model fit is poor. A reasonable conclusion might be that regional or local uniqueness, not global restructuring, lies at the heart of changing employment patterns.

Third, within the context of economic policy implementation, policy-makers are frequently overwhelmed by the sheer volume of information available. There exists, therefore, a need for quick, inexpensive analysis tools that are neither mathematically complex nor data intensive. Shift-share analysis is one such tool. Probabilistic shift-share provides a sound statistical basis for shift-share analysis. Since policy outcomes are, in part, dependent on the quality of information available, improving the quality of tools through which policy-makers filter information should lead to improved policy [63–66].

While the results presented here must be treated with caution, there also appears to be support for further use of information-theoretic models in the analysis of shift-share relationships. In this regard, such models have three positive characteristics. First, they provide a flexible vehicle for the derivation of shift-share models. Second, casting shift-share in information-theoretic terms places the methodology squarely into the mainstream statistical analysis of nominal, tabular data. Additionally, information-theoretic models have properties that are particularly useful for the analysis of selected forms of data. The information-theoretic form used here appears to be especially useful for analyzing data characterized by a small number of relatively large outliers. This issue is treated more fully by Flowerdew and Aitkin [53]. These advantages, however, must be weighed against the more difficult interpretation of information-theoretic models when used to analyze shift-share (see previous discussion and [67]).

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Appendix A

A1. Alternative calibration of the ANOVA shift-share model

An alternative calibration of the ANOVA shift-share model was devised by Patterson [33]. Patterson calibrates Eq. (4) subject to constraints that ensure that $\Sigma_i u_i B_i = 0$ for all i where u_i is the national employment in sector i. Should model (5) be used, then it must be the case that $\Sigma_i u_i B_i = 0$ for all i and $\Sigma_j v^j G^j = 0$ for all j where v^j is regional employment in j across all industries. This necessitates the use of constrained regression [68]. The resulting models are:

$$g_i^j = a + B_i + e_i^j \tag{4a}$$

subject to:

$$\Sigma_i u_i B_i = 0 \text{ for all } i$$

and

$$g_i^j = a + B_i + G^j + e_i^j \tag{5a}$$

subject to:

$$\Sigma_i u_i B_i = 0 \text{ for all } i$$

$$\Sigma_j v^j G^j = 0 \text{ for all } j$$

Results of an empirical comparison of models (4) and (4a) appear in Table 8. Values in column 1 of the table are generated by model (4), while those in the right column are generated by model (4a). It is evident from the table that model (4a) provides a less satisfactory explanation of the changes in US employment patterns over the study period than does model (4). Further, the differences in parameters between (4a) and (4) are not scalar in nature. Thus, different conclusions might be drawn from the two models.

Results of the empirical comparison of models (5) and (5a) appear in Table 9. In both

Table 8
Comparison of results from simple models

Parameter	Model (4)[a]	Model (4a)
Intercept	0.4206	0.4533
	(0.3947)	
Extractive industries	0.3765	0.0709
	(0.4024)	
Construction	−0.1013	−0.0575
	(−0.0754)	
Manufacturing	−0.2973	−0.2392
	(−0.2714)	
Transportation and utilities	−0.1848	−0.1512
	(−0.1588)	
Wholesale and retail trade	0.0564	0.0849
	(0.0823)	
F.I.R.E.	0.1121	0.1823
	(0.1380)	
Services	0.3955	0.3776
	(0.4214)	
Government	Aliased	−0.0052
	(0.0259)	
R^2	0.7078	0.4083

[a] Numbers in parentheses are not standard errors. See text for interpretation.

Table 9
Comparison of results from Arcelus extended models

Parameter	Model (5)[a]	Model (5a)
Intercept	0.4083	0.3959
	(0.3553)	
Extractive industries	0.2704	0.0709
	(0.2963)	
Construction	−0.1088	−0.0574
	(−0.0829)	
Manufacturing	−0.2796	−0.2392
	(−0.2537)	
Transportation and utilities	−0.1769	−0.1512
	(−0.1510)	
Wholesale and retail trade	0.0599	0.0849
	(0.0858)	
F.I.R.E.	0.1358	0.0182
	(0.1617)	
Services	0.4062	0.3776
	(0.4321)	
Government	Aliased	−0.0052
	(0.0259)	
New England	0.0476	−0.0500
	(0.0747)	
Middle Atlantic	−0.1241	−0.1553
	(−0.0970)	
South Atlantic	0.0367	0.0358
	(0.0638)	
East South Central	−0.0126	0.0237
	(0.0145)	
East North Central	−0.0069	0.0041
	(0.0202)	
West North Central	0.0190	−0.0022
	(0.0461)	
West South Central	0.2556	0.3316
	(0.2827)	
Mountain	0.2733	0.3253
	(0.3004)	
Pacific	Aliased	0.0033
	(0.0271)	
R^2	0.8670	0.7047

[a] Numbers in parentheses are not standard errors. See text for interpretation.

models, the intercept and parameters associated with the manufacturing, transportation and utilities, F.I.R.E., and service sectors, and the Middle Atlantic, the West South Central and Mountain regions are significant. This indicates significant employment growth at the national level, and that employment in manufacturing and transportation and utilities grew at a significantly lower rate than did that in national employment (whose parameter is, again, zero by definition). Conversely, in both models, employment in the F.I.R.E. and service sectors

grew significantly faster than did national employment. Regionally, employment in the Middle Atlantic region grew at a significantly lower rate, while the West South Central and Mountain regions grew at a significantly greater rate than did the nation as a whole. However, as was the case for model (4), in model (5) the lower R^2 associated with model (5a) indicates that the model provides a slightly less satisfactory explanation of the changes in US employment patterns over the study period than does model (5).

Summarizing, while the results presented here must be treated with caution, there appears to be some support for suggesting that Berzeg's calibration method [31,32] is superior to that proposed by Patterson. Perhaps the most noticeable characteristic of Berzeg's approach is its superior explanatory power. On the other hand, it is worth noting that the two models measure slightly different things. Berzeg's formulation measures growth in relation to a baseline industry and region, while Patterson's formulation measures growth with respect to national averages. Patterson's formulation is the more conventional and is more consistent when extended to the Arcelus form, but this, we feel, does not justify its use given that it is less accurate and computationally more expensive.

References

[1] Glickman NJ, Glasmeier A. The international economy and the American South. In: Rodwin L, Sazanami H, editors. Deindustrialization and regional transformation: the experience of the United States. Boston: Unwin Hyman, 1989. p. 60–80.
[2] Harrison B, Kluver J. Reassessing the Massachusetts miracle. Environment and Planning A 1989;21:771–801.
[3] Markusen AR, Carlson V. Deindustrialization in the American Midwest: causes and responses. In: Rodwin L, Sazanami H, editors. Deindustrialization and regional transformation: the experience of the United States. Boston: Unwin Hyman, 1989. p. 29–59.
[4] Huff D, Sherr L. Measure for determining differential growth rates of markets. Journal of Market Geography 1967;4:391–5.
[5] Plane DA. The geographic components of change in a migration system. Geographical Analysis 1987;19:283–99.
[6] Plane DA, Rogerson P. U.S. migration pattern responses to the oil glut and recession of the early 1980s: an application of shift-share and causative matrix techniques. In: Congdon P, Batey P, editors. Advances in regional demography. London: Belhaven Press, 1989. p. 386–414.
[7] Barff RA, Knight III PL. The role of federal military spending in the timing of the New England employment turnaround. Papers of the Regional Science Association 1988;65:151–66.
[8] Casler SD. A theoretical context for shift and share analysis. Regional Studies 1989;23:43–8.
[9] Barff RA, Knight III PL. Dynamic shift-share analysis. Growth and Change 1988;19:1–10.
[10] Houston DB. The shift and share analysis of regional growth: a critique. Southern Economic Journal 1967;33:577–81.
[11] Richardson HW. The state of regional economics: a survey article. International Regional Science Review 1978;3:1–48.
[12] Buck TW. Shift-share analysis: a guide to regional policy? Regional Studies 1970;4:445–50.
[13] Holden DR, Swales JK, Nairn AGM. The repeated application of shift-share: a structural explanation of regional growth? Environment and Planning A 1987;19:1233–50.
[14] Stillwell FJB. Further thoughts on the shift and share approach. Regional Studies 1970;4:451–8.
[15] Hellman DA. Shift-share models as predictive tools. Growth and Change 1976;7:3–8.
[16] Kurre JA, Weller BR. Forecasting the local economy, using time-series and shift-share techniques. Environment and Planning A 1989;21:753–70.

[17] Stevens BH, Moore CL. A critical review of the shift-share as a forecasting technique. Journal of Regional Science 1980;20:419–37.
[18] Rigby DL, Anderson WP. Employment change, growth and productivity in Canadian manufacturing: an extension and application of shift-share analysis. Canadian Journal of Regional Science/Revue Canadienne des Sciences Regionales 1993;26:69–88.
[19] Haynes KE, Dinc M. Productivity change in manufacturing regions: a multifactor/shift-share approach. Growth and Change 1997;28:150–70.
[20] Creamer DB. Industrial location and natural resources. Washington, DC: US Natural Resources Planning Board, 1943.
[21] Dunn Jr ES. A statistical and analytical technique for regional analysis. Papers of the Regional Science Association 1960;6:97–112.
[22] Perloff HS, et al. Regions, resources and economic growth. Baltimore: Johns Hopkins University Press, 1960.
[23] Arcelus FJ. An extension of shift-share analysis. Growth and Change 1984;15:3–8.
[24] Esteban-Marquillas JM. Shift- and share-analysis revisited. Regional and Urban Economics 1972;2:249–61.
[25] Thirlwall AP. A measure of the proper distribution of industry. Oxford Economic Papers 1967;19:46–58.
[26] Brown HJ. Shift-share projections of regional growth: An empirical test. Journal of Regional Science 1969;9:1–18.
[27] Browne LE. The impact of industry mix on New England's economic growth since 1950. New England Economic Indicators 1977;April:33.
[28] Fothergill S, Gudgin G. In defense of shift-share. Urban Studies 1979;16:309–19.
[29] Jackson G, et al. Regional diversity: growth in the United States, 1960–1990. Boston: Auburn House, 1981.
[30] Lever WF. The inner-city employment problem in Great Britian, 1952–76: a shift-share approach. In: Rees J, et al., editors. Industrial location and regional systems. New York: Bergin, 1981.
[31] Berzeg K. The empirical content of shift-share analysis. Journal of Regional Science 1978;18:463–9.
[32] Berzeg K. A note on statistical approaches to shift-share analysis. Journal of Regional Science 1984;24:277–85.
[33] Patterson M. A note on the formulation of a full-analogue regression model of the shift-share method. Journal of Regional Science 1991;31:211–6.
[34] Bureau of Labor Statistics. Establishment survey of employment, hours, and earnings. Washington, DC: US Government Printing Office, n.d.
[35] Kullback S, Leibler RA. On information and sufficiency. Annals of Mathematical Statistics 1951;22:78–86.
[36] Brockett PL, Charnes A, Cooper WW. M.D.I. Estimation via unconstrained convex programming. Austin, TX: Center for Cybernetic Studies, 1978.
[37] Knudsen DC. Exploring flow system change: U.S. rail freight flows, 1972–1981. Annals of the Association of American Geographers 1985;75:539–51.
[38] Fiacco AV, McCormick GP. Nonlinear programming: sequential unconstrained minimization techniques. New York: Wiley, 1968.
[39] Phillips FY. A guide to MDI statistics for planning and management model-building. Austin: Institute for Constructive Capitalism, 1981.
[40] Williams PA, Fotheringham AS. The calibration of spatial interaction models by maximum likelihood estimation with program SIMODEL. Bloomington, IN: Indiana University, Department of Geography (Geographic Monograph Series), 1984.
[41] Wilson AG. Entropy in urban and regional modelling. London: Pion, 1970.
[42] Feinberg SE. Analysis of cross-classified data. Cambridge, MA: MIT Press, 1981.
[43] Haynes KE, Machunda ZB. Considerations in extending shift-share analysis. Growth and Change 1987;18:69–78.
[44] Haynes KE, Machunda ZB. Decomposition of change in spatial employment concentration: an information-theoretic extension of shift-share analysis. Papers of the Regional Science Association 1988;65:101–13.
[45] Theil H, Gosh R. A comparison of shift-share and the RAS adjustment. Regional Science and Urban Economics 1980;10:175–80.
[46] Berry BJL, Schwind PJ. Information and entropy in migrant flows. Geographical Analysis 1969;1:5–14.
[47] Haynes KE, Phillips FY. Constrained minimum discrimination information: a unifying tool for modeling spatial and individual choice behavior. Environment and Planning A 1982;14:1341–54.

[48] Kullback S. Information and statistics. New York: Wiley, 1959.
[49] Thomas RW. Information statistics in geography. Norwich: GeoAbstracts, 1981.
[50] Knudsen DC. Generalizing Poisson regression: including a priori information using the method of offsets. The Professional Geographer 1992;44:202–8.
[51] Baxter M. Similarities in methods of estimating spatial interaction models. Geographical Analysis 1982;14:267–72.
[52] Baxter M. A note on the estimation of a nonlinear migration model using GLIM. Geographical Analysis 1984;16:282–6.
[53] Flowerdew R, Aitkin M. A method of fitting the gravity model based on the Poisson distribution. Journal of Regional Science 1982;22:191–202.
[54] Flowerdew R, Lovett A. Fitting constrained Poisson regression models to interurban migration flows. Geographical Analysis 1988;20:297–307.
[55] Flowerdew R, Lovett A. Compound and generalized Poisson models for interurban migration. In: Congdon P, Batey P, editors. Advances in regional demography. London: Belhaven Press, 1989. p. 246–56.
[56] Gokhale DV, Kullback S. The information in contingency tables. New York: Marcel Dekker, 1978.
[57] Lovett A, Flowerdew R. Analysis of count data using Poisson regression. Professional Geographer 1989;41:190–8.
[58] Lovett A, Whyte ID, Whyte KA. Poisson regression analysis and migration fields: the example of the apprenticeship records of Edinburgh in the seventeenth and eighteenth centuries. Transactions of the Institute of British Geographers, New Series 1985;10:317–32.
[59] McCullagh P, Nelder JA. Generalized linear models. London: Chapman & Hall, 1983.
[60] Knudsen DC. Manufacturing employment change in the American Midwest, 1977–86. Environment and Planning A 1992;24:1303–16.
[61] Knudsen DC, Barff R. Shift-share as a linear model. Environment and Planning A 1991;23:421–31.
[62] Massey DB, Meegan R. The anatomy of job loss: the how, why, and where of employment decline. London: Methuen, London, 1982.
[63] Murgatroyd L, Urry J. The class and gender restructuring of the Lancaster economy, 1950–1980. In: Localities, class and gender. London: The Lancaster Regionalism Group, Pion, 1985. p. 30–53.
[64] Brewer GD, deLeon P. The foundations of policy analysis. Homewood, IL: Dorsey Press, 1983.
[65] Laswell HD. A preview of policy sciences. New York: Elsevier, 1971.
[66] Weimer DL, Vining AR. Policy analysis: concepts and practice. Englewood Cliffs, NJ: Prentice Hall, 1989.
[67] Theil H, Sorooshian C. Components of the change in regional inequality. Economics Letters 1979;4:191–3.
[68] Pringle RM, Raynor AA. Generalised inverse matrices with applications to statistics. New York: Hafner, 1971.